

PRIME PURITY - STS OPERATING MANUAL

1.0

Introduction and General Principles

Purpose
Scope
At Sea
Risk Management

1.1

Control of Operations

General
Safe Watchkeeping
Manning for Transfer Operations
Propulsion Availability

1.2

Duties of Personnel

Person in Overall Advisory Control
Training
Security
Helicopter Operations

2.0

Conditions and Requirements

Ship Compatibility
Approval from Authorities
Transfer Area

2.1

Weather Conditions

Weather Forecasts
Weather Limits
Electrical Storms
Sloshing

3.0

Safety

General
Checklists
Safety During Cargo Transfer
Cargo Leakage
Deck Lighting

3.1

Emergency Shutdown Systems

Linked Emergency Shutdown Systems
Compatibility, Connection and Use
Emergency Release System
Electrical Isolation
Testing of Emergency Shutdown Systems
Action in Case of Infringement of Safety

4.0

Communications

Before Transfer
Notification
Information Exchange
Ship to Ship Transfer Checklists
During Transfer
Language
Navigational Warnings
Communication Failure and Emergency Signals

5.0

Mooring

Preparations Before Manoeuvring

5.1

Manoeuvring and Mooring

General
Controlling the Two Vessels Side to Side
Manoeuvring Alongside

5.2

Procedures Alongside

Under Way Transfer

Illustrations

5.0.1a

Typical Mooring Arrangement - 174,000m3 ~ 80,000m3

5.0.1b

Typical Mooring Arrangement - 174,000m3 ~ 174,000m3

5.0.1c

Typical Mooring Arrangement - 174,000m3 ~ 265,000m3

6.0

Cargo Transfer Operations

General
Cargo Tank Pressure
Manifold Condition
Cooling Down of the Cargo Lines
Cargo Tank Temperature

6.1

Hose Connections

Transfer Equipment
Electrical Insulation
Hose Connecting

6.2

Cargo Operations

General
Cargo and Ballast Sequence
Emergency Release Coupling and Hose Break System Testing
Custody Transfer Measurement

6.3

Cargo Tank Control

Equalise Tank Pressures
Cooling Down of the Cargo Lines
Ramping-Up and Ramping-Down
Boil-off Gas Control
Pressures Limit Guidance
Boil-off Management Equipment Failure
Maximum Levels

6.4

Hose Disconnection

Hose Draining and Purging
Hose Disconnecting

PRIME PURITY - STS OPERATING MANUAL

- 7.0
- Unmooring

Unmooring Procedures
Unmooring While One Vessel is at Anchor
Unmooring After Under Way Transfer
Unmooring Checks
Side-to-Side Operations
Unmooring Operation
Unmooring Using Quick-Release Apparatus or Toggle Pins

- 8.0
- Transfer Equipment

Fenders
Primary Fenders
Secondary Fenders

- 8.1
- Cargo Transfer Equipment

Introduction
Connection Pieces and Reducers
Hose Saddles
Cargo Transfer Hoses
Insulation Flanges for Transfer Hoses
Emergency Release Coupling
Cargo Transfer Hard Arms

- Illustrations
- 8.1.1a

Typical Transfer Hose and Y Piece Arrangement
- 8.1.1b

Insulation Flange Arrangement

- 8.2
- Mooring Equipment

Moorings
Ancillary Equipment

- 8.3
- Personnel Transfers

General

- 9.0
- Emergency Planning

General
Mooring Failure
Adverse Weather

- 9.1
- Emergency Disconnection

General
Emergency Escape
Ice Accumulation
Dead-Ship Emergency Release System Activation

1.0 Introduction and General Principles

1.1 Control of Operations

1.2 Duties of Personnel

PRIME PURITY - STS OPERATING MANUAL

1.0 INTRODUCTION AND GENERAL PRINCIPLES

Purpose

The purpose of this document is to provide guidance to the Masters and operators of vessels undertaking side-by-side Ship to Ship (STS) transfer, or lightering, of Liquefied Natural Gas (LNG). It is recommended that the guidance provided is followed by all vessels covered under the scope of this document.

Scope

This guide directly addresses the side-by-side STS transfer of LNG between commercially trading LNG carriers at anchor, alongside a shore jetty or while drifting. Tandem configuration alongside is not covered by this document. The procedures set forward in this Ship-to-Ship Transfer Operating Manual have been drawn up primarily in function of GTT No.96 and Mark III conventional membrane LNG carriers (LNGC) and the Q-Flex and Q-Max class.

At Sea

The term ‘at sea’ is used throughout this guide and is intended to indicate both the generally non-benign offshore waters and the sheltered waters, which are usually considered to be benign. It should be noted that a STS transfer operation ‘at sea’ may be conducted within the jurisdiction of a local (port) authority or national government. In such cases, reference should be made to applicable regulations and the possible need to obtain local approval for the activity.

Risk Management

A LNG STS transfer operation is an operation of some complexity that, at the time of compiling these guidelines, does not have the benefit of extensive industry experience as is the case with oil and LPG transfers. It is essential that the unique elements of each transfer be recognised and that a full and thorough risk assessment is performed as part of the planning of the STS transfer. The risk assessment should cover both the physical and operational hazards, and the means by which they are managed, and should also address the suitability of equipment and procedures.

The risk associated with any activity is a product of the likelihood (frequency) of any adverse occurrence, and the impact of such an adverse effect (consequence). Most normal activities involve a range of potential risks ranging from frequent, low-impact risks to rare high-impact risks. The consequences resulting from a LNG spillage during a STS transfer operation are potentially serious and effective mitigation measures should be put in place to reduce the likelihood of such an event occurring.

The risk assessment should identify all potential sources and consequences of risk for each particular operation. For some hazards, multiple risk events may exist, each with their own potential health and safety impacts and associated frequencies/probabilities. The risks identified will vary according to the type of STS operation under review, for example, double banked to a conventionally moored LNG carrier to Floating Storage Re-Gas Unit (FSRU); side by side with a LNGC at anchor or side by side with both LNG vessels drifting.

The risk assessment should take into account the risk reduction measures that are in place, their effectiveness and other factors that could change the probability/frequency of a risk event or its impact. Where administrative or procedural controls alone form the basis for risk reduction plans, these procedures should be thoroughly evaluated.

On completion of the exercise, significant risks will have been identified and an appropriate risk management strategy developed, with the inclusion of additional mitigation measures, as necessary, to ensure that all identified risks are reduced to an acceptable level.

Risk assessments are a part of the pre-transfer procedure and, as a minimum, will consider the following:

- Vessel compatibility.
- Location.
- Weather and sea state criteria.
- Personnel.

- Equipment.
- Environment.
- Emergency planning.

The risk assessment for a specific STS operation should be reviewed and revised as necessary in the event of any subsequent changes to operational parameters, equipment and procedures.

1.1 CONTROL OF OPERATIONS

General

The STS transfer operations will place additional demands on ships' crews, as personnel are not only required for the cargo transfer operations but may also need to keep a safe navigational or anchor watch throughout the operation.

Safe Watchkeeping

Prior to the commencement of a STS transfer operation, the person in overall advisory control (POAC), along with the Master of each vessel, should engage in a conference covering each key component of the operation (fendering, manoeuvring, mooring, hose connection, cargo transfer, hose disconnection, unmooring and unfendering). The purpose of the conference is to ensure that key personnel involved in the operation clearly understand and agree how the operation will be conducted.

Where the STS transfer operation is to be undertaken at sea, passage plans should include the consultation of relevant navigational charts and publications. The passage plan should detail:

- The projected track lines for the approach phase to the mooring operation.
- The required sea room.
- Planned anchor position.
- Anticipated track lines if drifting.
- Potential navigational hazards.
- Awareness of territorial boundaries (if applicable) and prohibited zones.

A proper deck and bridge watch should be established and maintained on each vessel drifting or at anchor for the duration of the operation.

Manning for Transfer Operations

In the planning phase for an LNG STS transfer operation, in co-operation with the Masters of the nominated vessels, due account shall be given to the estimated duration of the operation and an assessment made for the additional work loads and specific risks the STS operation will create. The aim should be to ensure that personnel remain fatigue free and minimum rest periods, as required by legislation, are complied with.

When planning STS transfer operations, ship operators, STS service providers and Masters should bear in mind that statutory minimum manning requirements may not address simultaneous operations, such as cargo operations together with a navigation or anchor watch. Where deemed necessary, following a detailed risk assessment, additional personnel should be placed on board to assist with the STS transfer operations.

Experience has shown that officers in charge still tend to carry out work or tasks whilst co-ordinating certain activities. This is unacceptable, it is stressed that supervising activities cannot go hand in hand with other tasks. The situational awareness shall be kept at all times during STS operations.

In order to keep a high safety standard and prevent fatigue, it is of utmost importance that work and rest periods are carefully planned beforehand. A matrix showing resting hours for each crew member is very helpful. Crew observing operations out of personal interest during their resting period have to remain well clear of the area. Resting periods shall not be used for training purposes.

As a minimum, during the STS operation, the complement should include:

PRIME PURITY - STS OPERATING MANUAL

- Two officers assigned to cargo duties, for example, chief officer or gas/cargo engineer assisted by a deck officer.
- At sea or at anchor, the bridge should be manned in accordance with the requirements of the STCW Convention.

The engine control room to be manned, as required.

Propulsion Availability

The engines during the STS transfer operation must be available at all times. The condition of the propulsion of each ship shall be confirmed during the pre-transfer meeting.

In case of a STS with both vessels moored and under way, it will be usually the SS or otherwise the larger vessel, that powers all manoeuvres. Nevertheless, final practices shall be confirmed by operational assessment during planning phase. More information can be found in the ICS/OCIMF Ship to Ship Transfer Guide (Petroleum, Chemicals and Liquefied Gas).

1.2 DUTIES OF PERSONNEL

Person in Overall Advisory Control

Notwithstanding the fact that each Master is responsible for their own vessel, the POAC may be one of the Masters' of the LNG carriers involved, usually the vessel that carries the equipment, or a designated individual from a STS service provider. The POAC should be responsible for the following:

- Ensuring agreed STS operating procedures are followed and that the operation is conducted in compliance with all applicable regulatory requirements.
- Ensuring that all checklists are completed.
- Ensuring completion of a Declaration of Inspection and/or ship to ship safety checklist.
- Overseeing the placement of primary and secondary fenders.
- Conducting a pre-operations dialogue with the responsible officers of both vessels and any lightering support craft.
- Discussing current and forecasted weather and sea state conditions for the duration of the operation.
- Discussing passage planning and agreeing courses and speeds for mooring operations including tugs, if any.
- Ensuring agreement with the mooring plan.
- Reviewing and ensuring that the site-specific risk mitigations are in place.
- Overseeing vessel approach and manoeuvring alongside the ship to be lightered (optional, can also be local pilot or mooring master).
- Ensuring the safe connection of liquid and vapour transfer lines and associated emergency release systems (ERS).
- Ensuring that the transfer lines are purged and leak tested prior to transfer.
- Ensuring that the emergency shutdown (ESD) system is properly connected and tested.
- Monitoring cargo transfer rates and vapour management.
- Monitoring current and forecasted weather and sea state conditions throughout operation.
- Monitoring the mooring arrangement.
- Monitoring communications throughout the operation.
- Supervising the assistant mooring master, if present, on the ship to be lightered.
- Ensuring that cargo transfer lines are properly drained and purged.
- Overseeing disconnection of liquid and vapour lines.
- Confirming the unmooring sequence.

- Overseeing unmooring and the separation of vessels.
- Overseeing the return of primary fenders, secondary fenders and transfer equipment to the support craft, as applicable.
- Completing a post transfer report on the STS operation, including lessons learned for future STS transfers.

Training

Additional training documentation or familiarisation briefings should be given to ship personnel addressing all aspects of the planned STS transfer operation. Proper crew training is one of the basic elements in performing the STS in a safe and efficient manner.

Security

The requirements of the International Ship and Port Facility Security (ISPS) Code, together with any local requirements, should be adhered to throughout the operation, with each vessel remaining responsible for their own security at all times.

Prior to commencing the STS transfer operation, the following security issues should be discussed and any associated decisions recorded:

- Exchange of sufficient information to allow completion of a Declaration of Security.
- Agreement on how, and between whom, communications regarding security are to be made.
- Actions to be taken in the event of a breach of security, such as suspending operations and separating vessels.

Local regulations may impose exclusion zones around the STS transfer operation. Due consideration should be given to establishing safe distances to be maintained from other vessels, and actions to be taken if such distances are compromised.

Helicopter Operations

Helicopter operations should not be carried out during manoeuvring, mooring operations or cargo transfer.

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2.0 Conditions and Requirements

2.1 Weather Conditions

PRIME PURITY - STS OPERATING MANUAL

2.0 CONDITIONS AND REQUIREMENTS

Ship Compatibility

A full compatibility assessment shall be undertaken prior to confirming the STS operation. The purpose of the assessment is to confirm the suitability of the two vessels for the operation to be performed and to identify any aspects that may require particular management. It should be noted that the level of detailed analysis required in such an assessment, particularly where specific vessel classes have not previously undertaken such an operation together, is extensive. This assessment requires considerable resources and may take several weeks or even months to complete. Compatibility assessments should include but not be limited to:

- Vessel characteristics.
- Manifold arrangements.
- Manifold platform arrangement and structure.
- Cargo handling equipment (cranes).
- Mooring arrangements.
- Excursion limits/operating envelope of the cargo transfer system.
- Parallel mid-body and fendering.
- Gas-safe areas.
- Personnel transfer.
- ESD and communication systems.
- Contingency planning and emergency procedures.
- Cargo conditioning - temperatures and pressures in both vessels.
- Cargo transfer and ballast plans.
- Vapour management capacity of each vessel.

A compatibility questionnaire should be provided by the operator of the cargo receiving vessel.

Approval from Authorities

Reference should be made to any local and national regulations applicable in the proposed transfer area (eg, within territorial waters or within the Exclusive Economic Zone (EEZ) of a country) to determine the level of approval required to conduct a LNG STS transfer operation. Once approval has been granted, the appropriate authorities and government agencies should be notified of the intended STS transfer operation.

Transfer Area

Consideration should be given to stability and cargo management issues, on both vessels, for the intended operation.

2.1 WEATHER CONDITIONS

Weather conditions under which LNG STS transfers can take place may impose restrictions on the operation. These restrictions should be studied for each specific transfer location and the vessels involved, along with procedures and contingencies for weather forecasting and operational management and shall include:

- Visibility.
- Wind direction and velocity.
- Wave/swell height.
- Wave/swell period and direction.
- Predictability.

Weather conditions may impose limits on the STS operation in following domains:

- Restrictions resulting from design of cargo containment system (sloshing), during transfer and in case of emergency abort.
- Maximum conditions in respect of anchor holding capacity.
- Vessel's relative dynamic motions.
- Mooring and fender loads.
- General operational constraints.
- Changes in atmospheric pressure, if significant, might have important impact on cargo tank pressures.

During the pre-transfer meeting, the expected weather conditions shall be discussed, as well as the moment on which to abort the operation, taking into account the time needed by the LNGC to rearrange levels within the cargo tanks in order to comply with the permissible filling limits for seagoing conditions.

Weather Forecasts

The weather conditions and weather forecast are of greater importance to the outcome of the LNG STS transfer than during a conventional STS transfer.

Care must be taken that when the transfer commences, weather conditions will remain within the below described limits during the complete transfer. Both vessels have the obligation to track weather systems and monitor the forecasts by all means necessary (Metmanager, weather facsimile, NAVTEX, EGC, etc). In addition, the STS Service Provider and/or STS Organisers will provide meteorological data containing specific detailed forecasts on wave periods, wave height, wind speed, visibility and other meteorological phenomena which might have a significant impact on the safety and success of the STS operation.

The weather forecasts cover a period of four days and are sent to the vessels by electronic mail. The forecasts commence a minimum of two days prior to the anticipated start of operations and continue twice daily until all operations are completed.

Weather Limits

The governing consideration on proceeding with a LNG STS transfer operation is that the operations will be performed in calm or slight seas. In order to mitigate severe sloshing effects associated with Resonance condition, the regions of pronounced Maximum Motion and Resonance for particular filling shall not coincide.

Electrical Storms

The Masters shall make a joint assessment of the electrical storm activity in the immediate area of the STS location. Based on the nature of the electrical storm and likelihood of passage, Masters shall jointly decide whether to suspend or continue with cargo transfer operations. During time of stoppage, all cargo pumps shall be stopped, but draining and purging of cargo hoses will not be required. The use of a stripping pump may be considered to maintain the liquid lines and hoses cold.

PRIME PURITY - STS OPERATING MANUAL

Sloshing

‘Benign’ is taken to mean an environment (sea state) in which critical sloshing phenomena does not occur. One of the main concerns of the LNG STS transfer are the periods of time when the LNG carrier is exposed to potential for liquid sloshing damage to the cargo containment system when the cargo levels are within the ‘sloshing zones’ of the vessel. Sloshing can become a problem with any level outside the required limits as is specified in the vessel builder’s specifications.

Within the barred filling range, the impact of LNG sloshing due to the motion of the LNG carrier may result in excessive impacts on the primary containment barrier and insulation system and excessive stresses on the cargo pump support columns. Two different situations have to be considered:

- Emergency departure: Implementation of measures to mitigate sloshing.
- STS in offshore conditions: Envelope curve for the operations to be developed.

Mitigating measures to reduce the risk for sloshing impacts consist in adapting operations by optimising unloading sequence and filling levels; and in case of developing emergency situations, by adapting ship's course and/or speed, or using bow and stern thrusters to keep the relative wave heading in the +/- 30° range of the bow or stern. Procedures and guidance tools have been developed by GTT to assist the ship staff in the decision making process.

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3.0 Safety

3.1 Emergency Shutdown Systems

PRIME PURITY - STS OPERATING MANUAL

3.0 SAFETY

General

For all STS transfer operations, each Master remains at all times responsible for the safety of their respective vessel, crew, cargo and equipment, and should not permit safety to be impacted by the actions of others. Each Master should ensure that the procedures recommended in this manual are followed and that internationally accepted safety standards are maintained. In addition to the guidance contained in this document, reference should be made to the latest industry guidance documents for STS transfer operations.

Checklists

Checklists for STS transfer operations should be used for each phase of the STS transfer operation.

Safety During Cargo Transfer

Regular physical and local checks should be carried out during transfer, mainly for the mooring and transfer system.

Emergency procedures should be prepared and their effectiveness reviewed during drills held prior to operations. During the compatibility and preparation phase, organisers shall make sure that there is conformity in the way both vessels react to contingencies. Consideration should be given to:

- Procedures for raising the alarm.
- Suspension of operations during emergencies.
- Deployment of mooring gangs to stations.
- Emergency disconnection of cargo transfer lines.
- Preparation of engines for manoeuvring.
- Unmooring.

Risk assessments should be used as a tool to identify and document potential emergency scenarios that are not already covered by the vessel's emergency response plan.

Cargo Leakage

During cargo transfer it is important to properly protect the hulls and decks of the two vessels by means of a water curtain, extended deluge system and water baths or other heat sinks. Therefore, on both vessels an additional deluge system shall be used to protect the tank slope and deck underneath the manifold drip tray. An additional water bath on the main deck under the manifold, either temporary or permanent, shall be installed under the cargo manifold area. A portable deluge system may be one which is connected to the existing fire mains by means of a fire hose and international shore connection. The additional deluge system is designed to replicate the existing hull protection system in way of the ship's hull between the main deck and the manifold area.

The system is capable of providing an adequate deluge flow rate which shall be ascertained by throttling the hydrant connected to supply the deluge piping. It shall be determined visually that the tank slope is at all times covered by the water curtain and that the basin underneath the drip tray is filled with water. This shall be confirmed before cargo transfer operations begin and routinely checked as the transfer progresses.

The suitability of the additional deluge system is confirmed during the STS ship compatibility study.

In case of major leaks (eg, rupture, defects and major incidents) all cargo operations shall be suspended, either manually or by ESD actuation.

Deck Lighting

During STS transfers at night, adequate deck lighting should be utilised. Obviously, navigational lights and signals required by the International Rules for the Prevention of Collision at Sea should be displayed.

3.1 EMERGENCY SHUTDOWN SYSTEMS

Linked Emergency Shutdown Systems

Linked ESD systems are used in the LNG shipping industry and enable terminals and ships to execute a controlled rapid shutdown of the cargo transfer operation in abnormal or emergency situations. A linked ESD system should be used for LNG STS cargo transfer. As far as is practical, all vessels should conform to the basic philosophy for ship/shore ESD set out in the SIGTTO document 'ESD Arrangements and Linked Ship/Shore Systems for Liquefied Gas Carriers' during cargo transfer operations.

Compatibility, Connection and Use

A detailed assessment should be made of the compatibility of the ESD systems of the participating ships, including confirmation of connector pin/channel.

Electrical or optical systems offer near instantaneous activation of both ships' ESD systems under trip conditions, as well as providing high reliability and continuous connection between the two cargo control room 'hotline' telephones.

In all cases, the cause and effects of ESD system activation should be discussed and agreed. In addition, both vessels should discuss and agree actions in event of ESD activation, including vapour management procedures. ESD communication channels may need to be reconfigured to ensure system functionality.

Telephone links provided via the appropriate ESD link may be incorporated into Communications planning discussions, as appropriate.

Emergency Release System

An Emergency Release System (ERS), consistent with 'dry-break' philosophies, is provided in accordance with the following guidance:

Each transfer line is fitted with an emergency release coupling (ERC) activated by a HPU.

The ERC is designed utilising two valves, one upstream and one downstream of the coupling.

The HPU is capable of operation and releasing all transfer lines in the event of ship blackout and the non-availability of ship-provided services.

In all cases, ESD 1 and ESD 2 are interlocked such that ESD 2 is not possible without ESD 1 taking place first.

A system is in place to ensure activation of any ERC is not initiated before the emergency shutdown (ESD1) trip signal has been issued on both ships.

In the event of the ships breaking away, the HPU is automatically operated and releases the transfer lines ERCs before the maximum operating limits of hoses are reached. Equipment is also in place to enable manual activation of the ERCs.

The system is designed such that operation will not result in unacceptable surge pressure in vessel piping. Refer to Guidance for the Alleviation of Excessive Surge Pressures on ESD (SIGTTO), for further information.

Full consideration is given to the hose ends upon an emergency release with the avoidance of sparking a key objective.

Systems are designed and procedures developed such as to minimise the possibility of spurious operation.

Details on the ERS can be found in the relevant sections of the Hose Handling Manual.

PRIME PURITY - STS OPERATING MANUAL

Electrical Isolation

At present, inter-vessel electrical isolation between the ships is maintained during transfer line connection/disconnection and cargo transfer operations. This is to reduce the risk of high-energy sparks being produced due to the electrical potential difference between the hulls.

Attention should be paid to the possibility of radio-frequency induction when cranes are used, particularly when handling electrically conducting hoses. The supporting steel deck, crane structure, lifting wires, shackles, and hose can form an open-ended induction loop and may lead to arcing between the hose end and the steel deck or other part of the ship's structure. Main MF/HF radio transmitters should be switched off and the antennae earthed during hose handling and cargo transfer operations.

As a minimum requirement, each cargo transfer line has an electrical isolation arrangement in the form of an insulation flange. The insulation flange is installed on the end of the cargo transfer hose that is connected to the emergency release coupling, ensuring equal potential of the ship to be lightered (STBL) and the hose after emergency release. A hose bonding wire is used during connection and disconnection. All hoses should be electrically conducting.

Testing of Emergency Shutdown Systems

Both ships should test their emergency shutdown systems not more than 48 hours before STS operations commence and should advise the POAC of successful completion of these tests. The tests should be documented in accordance with the STS procedures and checklists.

Before STS cargo transfer commences, the ESD systems on both vessels should be re-tested to ensure correct system operation and to prove that ESD trip signals can be communicated between the ships. As per standard practice, both vessels will arrive with their ESD systems tested and automated valves closure times adjusted.

The SIGTTO document 'Guidelines for the Alleviation of Excessive Surge Pressure on ESD' states that a terminal will shut down first during loading of cargo and the ship will shut down first on discharging, this to protect transfer hoses from excessive pressure surges. These basic good practices which SIGTTO promotes are put into practice, adjusting the closing times of the ESD valves on both vessels, and as such, ESD valves on the discharging vessel should close faster (about 25 seconds) than ESD valves on the receiving vessel (about 30 seconds).

Once the hoses are connected and STBL is ready to start the cooling down, she will initiate an ESD in warm condition. The SS will afterwards perform a cold ESD. During ESD testing, the vessels may have to remain in Dual Fuel mode.

The mechanical release mechanism of the ERS should be proven operational and ready for use before cargo transfer commences. Tests of the system should prove the correct operation of all components, with the exception of actually releasing the ERC.

The system that ensures that ESD 1 occurs prior to ESD 2 should be tested and proven operational.

The actual physical operation of the ERC to verify separation should be tested once per year, or as per manufacturer's recommendations, whichever is the more frequent.

Action in Case of Infringement of Safety

If either ship fails to observe any of the safety requirements during the STS transfer operation, the infringement should be brought to the attention of the Master of the ship concerned and the POAC. Operations should be suspended, as appropriate, until the situation is rectified.

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4.0 Communications

DRAFT MANUAL FOR REVIEW & COMMENT

PRIME PURITY - STS OPERATING MANUAL

4.0 COMMUNICATIONS

Before Transfer

An essential requirement for successful STS transfer operations is the maintenance of good communications between the involved ships. The ships should establish initial communication as early as practicable to plan operations and to confirm the transfer area.

The authority of the POAC has to be mutually agreed upon and relevant logbook entries to be made.

As communication routines involving authorities are site dependent, parties involved should liaise with STS Service Providers and local agents for obtaining precise and updated requirements on information exchange procedures.

Notification

The ship to be lightered (STBL) is to notify the service ship (SS) upon sailing from her load port and relay her ETA at the STS position. In the notification, a copy of the cargo quality certificate, the liquid temperature and cargo tank pressures should be included. On a daily basis, the vessel needs to provide an update on the average liquid temperature and absolute vapour pressure of the cargo tanks. The ETA is to be updated 96 hours, 48 hours, 24 hours and 12 hours before arrival. The 24 hours message is to include any known defects to the vessel and all necessary information, as well as all performed tests. Receipt of this message is to be confirmed by the SS and same information to be transmitted to the STBL.

Information Exchange

An information exchange, including STS questionnaire and checklists, shall take place well beforehand between STS Service Provider and both vessels. Latest two days prior to commencement of operations, STS Service Providers or agents shall forward weather forecast to parties involved. For practical organisational details, communication with lightering support vessel should be established if required.

Ship to Ship Transfer Checklists

Communication systems and back-up means are confirmed during the pre-transfer meeting. Relevant STS checklists have to be satisfactorily completed and signed by the two vessels prior to proceeding into the next level of the operation. In US waters, the Declaration of Inspection (DOI) should be completed as well, this is a documentary requirement stipulated in the US Code of Federal Regulations. Additional paperwork and timesheets shall also be discussed during the pre-transfer meeting.

More information can be found in the ICS/OCIMF Ship to Ship Transfer Guide (Petroleum, Chemicals and Liquefied Gas).

The approach should not be initiated before both Masters and the POAC have agreed all the procedures and intended courses and speeds.

Checklist 3 must be completed prior to commencing cargo transfer. The relevant authorities should be informed of the anticipated time of unmooring.

During Transfer

Radio frequencies to be compared between both vessels and a working channel to be agreed. A set of two UHF portable radios with spare battery should be available to transfer to the STBL in case needed. On both vessels, the cargo watches have to be maintained in compliance to STCW and ILO regulations concerning rest hours. During the transfer, cargo statistics are to be exchanged, logged between vessels at the agreed intervals (usually hourly).

Primary means of communication are the UHF portable radios. Secondary means of communication is the hot-line telephone. The hot-line phone transmittal happens over the electrical SSL umbilical. The primary and secondary means of communication shall be tested prior to start of operations.

Furthermore, the bridge watches have to maintain and monitor common VHF channels during the complete STS operation. The time log has to be maintained by the bridge watchkeeping officers as well as in the cargo control room (CCR) for all the events.

Language

The command language between vessels engaged in STS operations has been defined as English.

Should a problem be experienced with communications not being understood, operations should be suspended until the matter is resolved.

Navigational Warnings

Prior to commencing a STS transfer operation, the recommendations contained in the appropriate section of the ICS/OCIMF Ship to Ship Transfer Guide (Petroleum, Chemicals and Liquefied Gas) should be followed with respect to the issue of navigational warnings.

Communication Failure and Emergency Signals

Should there be a communications failure, then all operations shall be suspended until corrective action has been undertaken to the satisfaction of POAC and Masters.

The emergency signal should be tested before arrival with both vessels agreeing on a mutual emergency signal (eg, 5 or more short and rapid blasts on the ship's whistle) and the response to it.

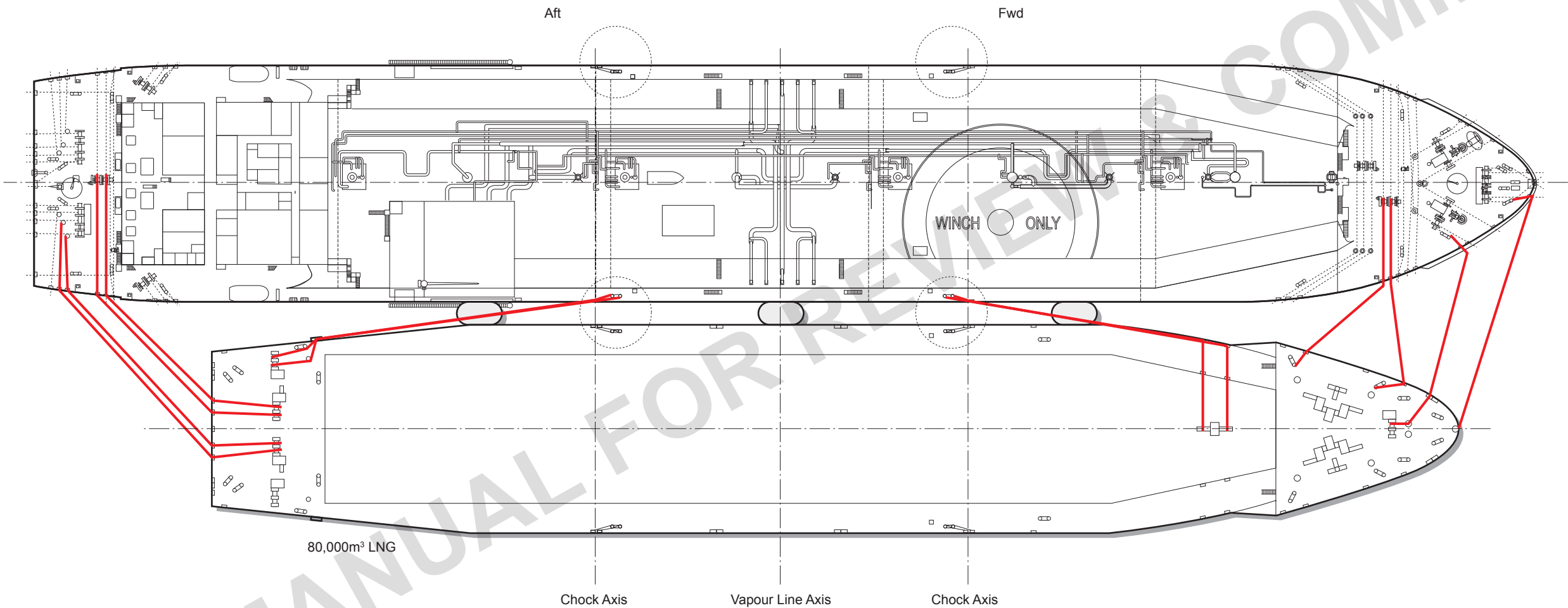
5.0 Mooring

5.1 Manoeuvring and Mooring

5.2 Procedures Alongside

PRIME PURITY - STS OPERATING MANUAL

Illustration 5.0.1a Typical Mooring Arrangement - 174,000m³ ~ 80,000m³



PRIME PURITY - STS OPERATING MANUAL

5.0 MOORING

Preparations Before Manoeuvring

Reference should be made to ICS/OCIMF Vessel to Vessel Transfer Guide (Petroleum, Chemicals and Liquefied Gas) for information on operational preparations to be undertaken on board both vessels prior to manoeuvring.

Prior to the STS operation, the Masters of both vessels and, if appointed, the POAC, should make the following preparations before manoeuvres begin:

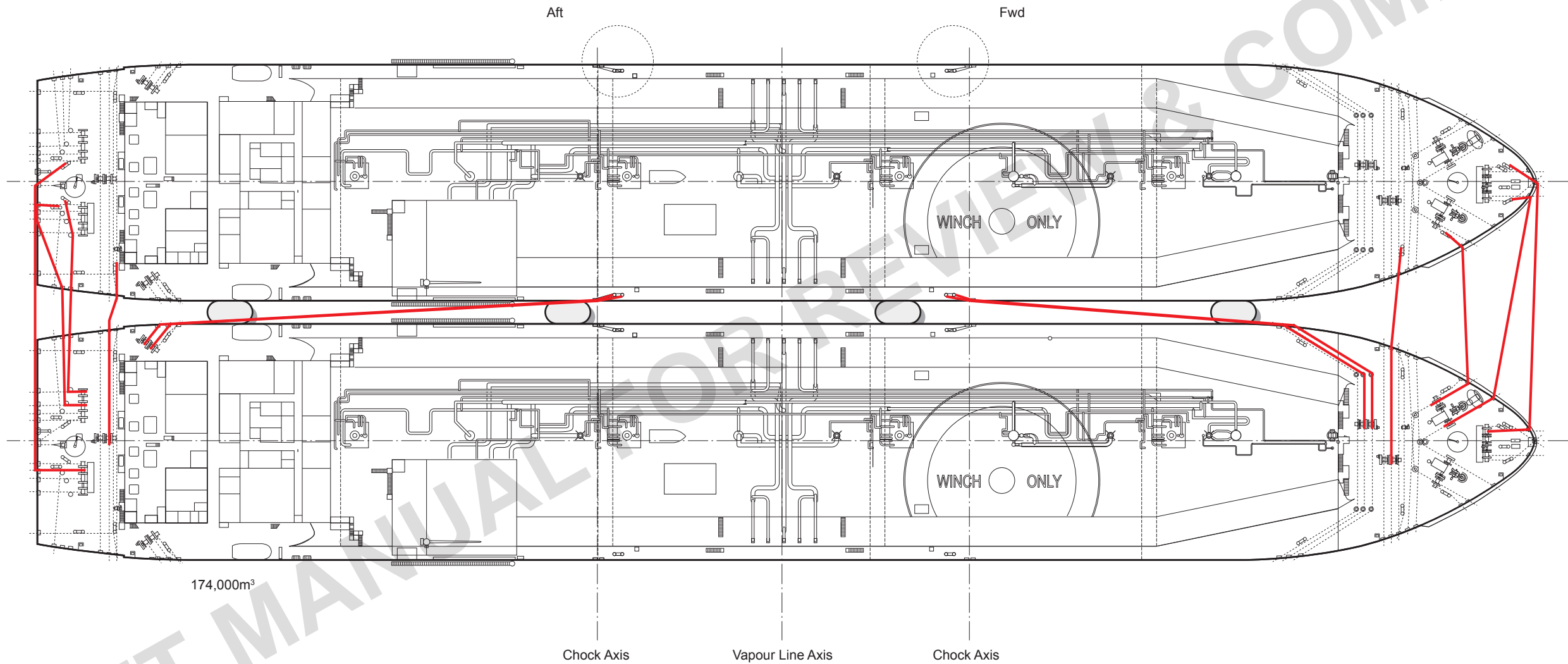
- Carefully study the operational guidelines contained herein and in the industry publication ICS/OCIMF Vessel to Vessel Transfer Guide (Petroleum, Chemicals and Liquefied Gas), as well as any additional guidelines provided by the vessel owner and cargo owner.
- Ensure that the crew is fully briefed on procedures and hazards, with particular reference to mooring and unmooring.
- Ensure that the vessel conforms to relevant guidelines, is upright and at a suitable trim.
- Confirm that the steering gear and all navigation and communications equipment is in satisfactory working order.
- Confirm that engine controls have been tested and the main propulsion plant has been tested ahead and astern.
- Confirm that all essential cargo and safety equipment has been tested.
- Confirm that mooring equipment is prepared in accordance with the mooring plan.
- Fenders and transfer hoses are correctly positioned, connected and secured.
- Cargo manifolds and hose handling equipment is prepared.
- Obtain a weather forecast for the STS transfer area for the anticipated period of the operation.
- Agree the actions to be taken if the emergency signal is sounded.
- Confirm completion of relevant pre-operational checklists.
- Confirm the Security Level at which the vessels are operating in accordance with the provisions of the ISPS Code, and the requirements being undertaken on board to ensure compliance. When a vessel is conducting vessel to vessel activities with another vessel not required to have or implement an approved vessel security plan, a Declaration of Security should be made (ISPS Code Part).
- The Navigational Signals to be shown and the sound signals made, during STS transfer operations are those required by the International Regulations for Preventing Collisions at Sea (COLREGS).

Communications with the Master of the other vessel should be established in accordance to co-ordinate the rendezvous and the method and system of approach, mooring and disengaging.

When the preparation of either vessel has been completed, the other vessel should be so informed. The operation may proceed only when both vessels have confirmed their readiness.

PRIME PURITY - STS OPERATING MANUAL

Illustration 5.0.1b Typical Mooring Arrangement - 174,000m³ ~ 174,000m³



PRIME PURITY - STS OPERATING MANUAL

5.1 MANOEUVRING AND MOORING

General

In side to side transfer, one vessel (the constant heading vessel), normally the larger, maintains steerage way at slow speed (preferably less than 5 knots) on a constant heading. Normally, the manoeuvre will be made with the wind and sea ahead, however, local conditions and knowledge may dictate an alternative heading. The manoeuvring vessel then manoeuvres alongside.

It is recommended that the manoeuvring vessel approaches and berths with her port side to the starboard side of the constant heading vessel.

Controlling the Two Vessels Side to Side

Each vessel should take recommendations reflected in the ICS/OCIMF Vessel to Vessel Transfer Guide (Petroleum, Chemicals and Liquefied Gas) and following into account:

- Engine controls, steering gear and all navigation and communications equipment should be in full working order.
- Proficient helmsmen should be assigned to steer each vessel.
- Courses requested by the manoeuvring vessel must be followed by the constant heading vessel.
- Vessel's speed should be controlled by adjusting engine revolutions (or propeller pitch). Any adjustment should be limited, for example, to plus or minus 5 rpm rather than using the relatively coarse engine room telegraph system. However, normal full operating range must remain readily available.
- At night, the deck should be adequately lit and, if possible, the vessel's side and fenders should be spotlighted.
- The side for mooring should be clear of all over-side obstructions, permanent and otherwise. In case it is not possible to create such conditions, extreme precautions should be taken in the course of STS operations, keeping the mooring lines tight.
- The navigation lights and shapes appropriate to STS transfers should be displayed.
- There should be effective radio communications between the bridge and mooring gangs.
- There should be effective communication between the Masters of each vessel.

Manoeuvring Alongside

Although individual Masters will have their own preference for the method of manoeuvring their vessel, the following points are emphasised for STS transfer operations:

Mooring winches are to be considered as the basic means to heave-up lines but not as the means to pull vessel to any object of mooring.

- The Masters of both vessels should always be prepared to abort a berthing operation if necessary. The decision should be taken in ample time while the situation is still under a control.
- The effect on both vessels when using twin screws during close manoeuvres.
- Common method of berthing is for the manoeuvring vessel to approach the constant heading vessel from the quarter on the side of berthing.
- At all times, each vessel is responsible for maintaining a proper lookout.
- Generally during manoeuvring, the wind and sea should be ahead or nearly ahead, however, local conditions or knowledge may indicate an alternative approach.
- The angle of approach adopted by the manoeuvring vessel should not be excessive.
- On closer approach, she should parallel the course of the constant heading vessel, about 50 to 100 metres off. The manoeuvring vessel should then position it relative to the constant heading vessel.
- The manoeuvring vessel reducing the distance by appropriate rudder and engine movements until fenders touch/make contact.

The two vessels should preferably make parallel contact at the same speed with no astern engine movements being necessary, and the effects of vessel interaction should be anticipated when manoeuvring at close quarters.

No mooring lines will be allowed in the way of the manifolds. Where possible, mooring lines shall be taken in pairs and mooring lines to be put preferably on the inboard bits.

The STBL should have buoyant rope messengers ready between dedicated fairleads towards winches and capstans. The SS can, if required, send heaving lines by line-throwing apparatus to pick up the rope messengers. Recommended messenger lines should be 40mm 8-strand polypropylene of braided construction.

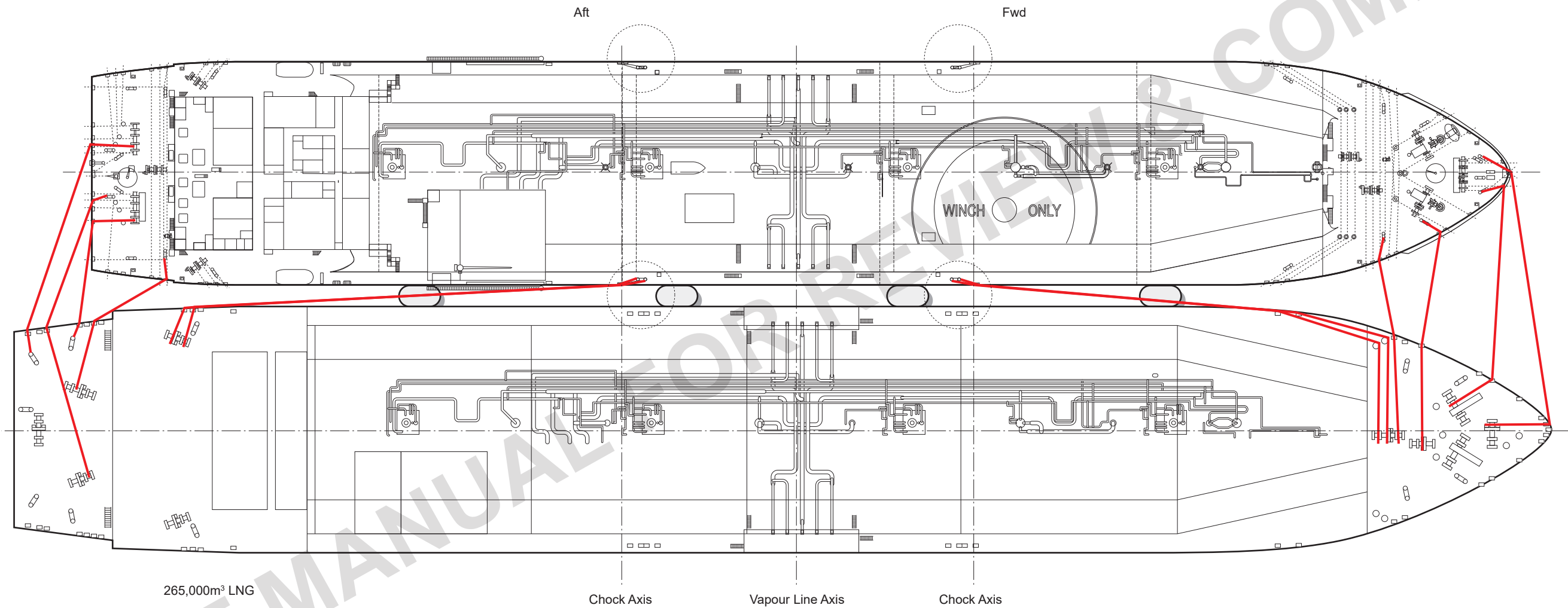
To facilitate mooring of the springs, synthetic ropes are commonly fitted between the spring wires and tails by means of a shackle or a “cow hitch”. These lightweight ropes seriously reduce the weight of the spring lines so that they can be easily handled, without reducing the breaking strength.

The length of the synthetic rope is chosen in such a way that the steel wire and synthetic tail will take all the chafing. When applying sufficient grease on the long synthetic tails, no damage from chafing should occur.

Toggle pin arrangement shall be used on all mooring bitts. The toggle pins allow for a rapid and safe releasing of lines, and the need to shift the toggle pin to the next mooring bitt would slow down the unmooring in case of emergency release.

PRIME PURITY - STS OPERATING MANUAL

Illustration 5.0.1c Typical Mooring Arrangement - 174,000m³ ~ 265,000m³



PRIME PURITY - STS OPERATING MANUAL

5.2 PROCEDURES ALONGSIDE

For procedures alongside, reference should also be made to Checklists 3 and 4. The LNG cargo transfer operation is subject to normal cargo planning processes prior to the operation and the plan should be agreed in writing between the two vessels.

The cargo plan should include, as a minimum:

- Quantity to be transferred, including quantities on board before and after transfer.
- Cargo and cargo tank temperatures.
- Estimated saturated vapour pressure of cargo to be transferred.
- Cooldown procedures.
- Initial transfer rate and maximum transfer rate.
- Vapour pressure differentials and maximum allowable values.
- Ballasting and deballasting arrangements taking into account relative freeboard.
- Custody transfer arrangements and documents.
- Procedures for increasing/reducing transfer rates.
- Procedures for the control of vapour pressures.
- Identification of POAC of the operation.
- Watch or shift arrangements.
- Critical stages of the operation.
- Local or government rules that apply to the transfer.
- ISPS security exchange.
- If applicable, sloshing zones and a contingency plan for internal cargo transfer in the event of a prolonged interruption of cargo transfer.

Under Way Transfer

Local conditions, such as those where water depths are too great for anchoring, sometimes demand that cargo transfer be carried out with the two-vessel system under power and making way through the water. As long as adequate sea room is available, traffic conditions, weather, sea conditions and forecasts are suitable, then transfers of this type can be carried out, but large diameter fenders of an acceptable type should be used. It should be noted that speeds through the water should be minimal, preferably in the 1 to 2 knot range.

The constant heading vessel maintains steerage way at slow speed on a steady course (approximately 1 to 2 knots) and the (former) manoeuvring vessel keeps its rudder amidships and remains (with engines stopped) as a towed vessel. In order to minimise towing loads on the moorings, the constant heading vessel should alter her engine revolutions sparingly, adjusting speed very gradually.

The chosen course and speed should be agreed by the two Masters and should result in minimum relative movement between the two vessels and minimum turbulence in the gap between the hulls.

Under such circumstances, while the vessels are moored together as a unit, safe navigation and collision avoidance is usually the responsibility of the constant heading vessel, but may be under the direction of the person in overall advisory control aboard the other vessel.

As an alternative and provided conditions are suitable and a transfer area of suitable size is available, it can be advantageous to carry out cargo transfer while the two vessels are allowed to drift freely.

The use of the under way transfer system requires a full navigational watch to be maintained on the bridge of each vessel.

If deteriorating weather conditions cause the vessels to roll unacceptably, engine movements can be used to bring the vessels to a heading, which results in minimum movement.

Constant attention should be paid to mooring lines and fenders to avoid chafing and undue stress, particularly that caused by changes in relative freeboard when fenders can easily jump out from the water onto the deck.

Mooring lines are to be monitored every time, and greasing or other protective means (various covers - such as old rubber hose or old fire hose) can be applied accordingly as protective sleeves.

If at any time mooring lines need to be repositioned or adjusted, this should only be done under strict controlled conditions.

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6.0 Cargo Transfer Operations

6.1 Hose Connections

6.2 Cargo Operations

6.3 Cargo Tank Control

6.4 Hose Disconnection

PRIME PURITY - STS OPERATING MANUAL

6.0 CARGO TRANSFER OPERATIONS

General

A pre-transfer meeting, preferably to take place on board of the STBL, is organised with representatives of ship's staff personnel (cargo officers of both vessels as a minimum). The Masters of both vessels can decide to have the meeting prior to starting the connection of the hoses. If both Masters agree, the connection can take place during the pre-transfer meeting. The 'Pre-Transfer Meeting Document' contains key elements of the STS procedures and covers any issues to be agreed on.

Cargo Tank Pressure

A very important aspect of the STS operation is the control of the cargo tank pressure. Experience has shown that during a STS cargo transfer, a rise in liquid temperature of between 0.6°C to 1.0°C can be expected. This rise in temperature comes from the heat added by the cargo pumps, weather conditions, sea conditions etc. For a some cargo types, a 1.0°C increase in temperature would correspond with a pressure increase of 10.0kPa. For a safe STS operation, the temperature of the cargo on the STBL should allow for this increase in pressure on the SS. The maximum saturated vapour pressure, after allowing for the rise in pressure, should not exceed 21kPa. This will allow a sufficient safety margin with respect to the safety relief valve setting in order to handle the cargo tank pressure, even in the event of a boiler management equipment failure.

Boil-off management systems such as steam dump, reliquefaction and GCU shall therefore be evaluated and confirmed operational at the planning phase.

Manifold Condition

Prior to arrival, the STBL shall assure that manifolds are properly purged with nitrogen and blind flanged with 4 bolts tight. Liquid manifold connections not to be used are to remain blinded and secured with all bolts/nuts secured tight.

Cooling Down of the Cargo Lines

Cooling down of the cargo lines shall only start after the hoses are connected and it is of utmost importance that no cooling down of the lines takes place during the hose connection. Prior to hose handling, the responsible cargo officer on the STBL will confirm to the SS that no pumps are running and that the lines are properly drained.

Note: When cooling down the cargo lines, they shall be cold up to the inboard manifold manual valve (90% frosted) having the forward and aft liquid header temperature at about -110°C.

Cargo Tank Temperature

The available cargo heel permitting, the SS will ideally arrive with the tanks cooled down (ATR -130°C) to normal (GTT) standards with liquid on the bottom of all four tanks. Barriers shall be filled adequately with nitrogen pressure; nitrogen flows stabilised and buffer tank pressurised above 6 bar (600kPa).

6.1 HOSE CONNECTIONS

Transfer Equipment

The SS shall have all cargo transfer equipment connected, leak tested, purged with nitrogen and fully ready for use upon arrival.

Electrical Insulation

Both vessels shall switch off their cathodic protection systems prior to arrival, this as part of a set of measures to eliminate the risk of electrical arcing during the connection and hose handling operations.

A hull bonding cable will NOT be connected.

Hose Connecting

Hose handling and connection will be guided by the chief officer of the SS, the vessel fitted with STS equipment, or his delegate. The POAC will oversee the connection and may provide corrective guidance as and if necessary. As the crew of the STBL is actively involved during connection, the SS shall be provided with a copy of the relevant documentation well in advance of the operation (a minimum of 2 weeks is required for study and training).

The hoses (8" -NB) and emergency release couplings (ERC) are used to connect the liquid and vapour manifolds. An adequate torque of 250-270Nm is to be applied to all connection flanges of the hoses and ERCs. Flange insulation sets shall be inserted on board the SS between each hose and the ERC.

When the SS arrives at the STS location, the hoses will already be under a nitrogen atmosphere and flanged outboard. During connection to the STBL, the blind flanges are taken off and therefore the SS will have to purge all the hoses with nitrogen until the other vessel measures oxygen content of less than 5%. For purging, the manifold purging valves to be used: at the SS to connect the nitrogen hose, the other ship to vent to the atmosphere and measure the oxygen content.

Prior to start, cooling down of the manifold and hoses, a leak test of all connection flanges with nitrogen shall take place. Test pressure for liquid lines is 450kPa (4.5 bar) and for vapour line is 150kPa (1.5 bar). All hoses have to be depressurised to 5-10kPa after the test.

PRIME PURITY - STS OPERATING MANUAL

6.2 CARGO OPERATIONS

General

The ship to ship pre-transfer meeting shall finalise the mutual compatibility issues on cargo transfer rates and emergency contingency planning. Subsequently, both vessels shall make simultaneously a clocks time check and be synchronised), and counters noted down. The emergency shutdown (ESD) system and ERC actuation system shall be tried and confirmed operational prior to starting the hose cooldown.

Tank pressures to be equalised prior to start of the cargo transfer. Whilst cooling down, the ESD system shall be kept activated. Cooling down is considered completed when the SS confirms a minimum temperature of -110°C or below on her lines.

The maximum total transfer rate will be between defined rates, subject to agreement. Orifices are used, except on the vapour hoses, to limit the flow through each hose in line with above agreement.

The cargo transfer during STS operations is a standard discharge/loading operation, except that special care has to be taken about following items:

- BOG control.
- Check of torques on manifold flanges before, during and after cooling down.
- Sequence (sloshing zones).
- Maximum filling levels.
- Topping-off rates.

Cargo and Ballast Sequence

It is of utmost importance that the various stages of the unloading/loading sequence are carefully planned. The prepared sequence shall be as such that the period of time when the cargo levels on board are inside the sloshing zones is limited to a strict minimum.

The various stages of the unloading sequence shall be discussed during the pre-transfer meeting, when both ships shall demonstrate compliance with the above and that intermediate steps are acceptable for stability and strength.

Emergency Release Coupling and Hose Break System Testing

The dry break ERC will be tested in warm condition only. The test shall be performed following the guidelines contained by detailed procedure for ERC handling and testing. The actuation pin will be removed before the test; a test in warm condition only is sufficient; moreover, re-inserting the pin when ERC is frozen may prove difficult. The hose brake pneumatic clutch system shall be tested for proper operation. Upon completion of the tests, the systems shall be reinstated and confirmed fully operational.

Custody Transfer Measurement

Gauging will preferably be done before starting the warm ESD and ERC tests. At the same time the ‘gas to engine’, ‘gas ashore’ and ‘gas from shore’ counters (if available) will be noted down. Upon completion of cargo transfer, after liquid draining, closing CTMS will be carried out and counters taken again. Float readings will be used as back-up for the remote automatic gauging system. Consideration should be given to sloshing of cargo liquids during gauging. Vessels should NOT secure gas burning prior to gauging. Witnessing parties shall be ascertained during the pre-transfer meeting and prior to CTM, clocks shall be synchronised.

Sampling may take place according to commercial instructions and standard safe working practices of each vessel.

PRIME PURITY - STS OPERATING MANUAL

6.3 CARGO TANK CONTROL

Equalise Tank Pressures

After CTM, tank pressures to be advised to either vessel prior opening vapour manifold. Tank pressure will be adjusted by throttling the vapour control valve. It is acceptable that the SS tank pressures are below the STBL tank pressures, as for effective STS flow rates, the tank pressures on receiving vessel may increase up to 60-70mbar compared to the laden vessel.

Cooling Down of the Cargo Lines

Upon completion of hose handling activities and warm ESD testing, both vessels will cool down their deck cargo lines. When the deck cargo lines are cooled down, the transfer hoses, spool pieces, ERCs and manifolds remain to be cooled down. Cooldown should be carried out slowly; time required for cooling down transfer hoses is usually 1.5 hours. A rushed cooldown will only increase the risk for leaks.

During cooldown, the STBL shall use one or two stripping/spray pumps depending on the head required. The coolant will be sent via the ESD valves (open) and flow rate will be regulated by the small bypass valves coming from the stripping header. The line-up on the STBL has to be such that during an ESD the actual stop of flow is initiated at the side of the manifold.

On the SS, the coolant will be passing the ESD valves and then into the bypass valves towards the stripping header. The LNG will then be flowing towards the spray line at the vapour dome till the spool piece on the SS is 50% (about -20°C) frosted.

On achieving the required temperature, the cooldown is stopped for a cold ESD test, after which, lining-up takes place for cargo transfer.

The detailed cooling down procedure will be agreed during the pre-transfer meeting. Both vessels will check the torque settings of all bolts at the manifold upon completion of the cooldown operation.

Ramping-Up and Ramping-Down

The actual transfer rate will very much depend on the condition of the LNG cargo on board the delivery vessel and the temperature of the cargo tanks on the receiving vessel.

Ramping-up rate is subject to the tank pressure of the receiving vessel. The ramping-up required rates and intervals shall be discussed during the pre-transfer meeting, when a ramp-up schedule shall be proposed by the delivery vessel. Normal practice is to keep the liquid flow during start-up of main cargo pump at a low rate of about 800m³/h. During this start-up period, careful checks will take place to timely spot any irregularities. The time from start-up of 1st cargo pump until full transfer rate is about 1 hour.

The ramping-down cargo transfer at required rates and intervals shall be discussed at the pre-transfer meeting.

Boil-off Gas Control

The vapour return during STS will be done in the free-flow method; it is not feasible to use any kind of compressor.

For an empty ship, it is of utmost importance that a level of liquid is maintained at the bottom so as to avoid the BOG flash during the initial phase of loading. During the transfer, there will be an increase of BOG due to the heat influx from the neighbouring barriers (insulation).

Both vessels will consume or re-liquefy gas to the maximum of their capabilities; any limitations on the vapour management capabilities shall be duly reported during pre-transfer meeting. If the vessel is unable to use its reliquefaction plant, then the full use of the GCU is required. Also, prior to the STS operation, both vessels confirm the good working condition of the BOG management equipment.

Vapour return pressure is controlled from the STBL by regulating the vapour valve. If this is not sufficient, the LNG transfer rate has to be decreased until the cargo tank pressure is under control.

As for any cargo operation, absolutely no vapour will be vented to the atmosphere. At all times, LNG transfer rates will be adjusted as a function of boil-off gas control.

Pressures Limit Guidance

Following guidance is given regarding pressure control during STS operations. The aim is to avoid having to vent vapour while the two vessels are alongside each other. Prior to the STS operation, the liquid temperature and related saturated vapour pressure on the STBL are checked. These parameters should be well within limits, to avoid reaching high cargo tank pressures in the SS.

Below pressure limits and related measures are in use by experienced industry players and may provide good guidance; nevertheless, final limitations will be decided by vessel operator based on proper pre-planning assessment:

17.5kPa: Measures to be taken to slow down the cargo transfer rate to keep the pressure under control.

19.5kPa: The cargo transfer must be stopped. The operations can only be resumed when the cargo tank pressures can be kept under 17.5kPa.

20.5kPa: Consideration should be made for the vessel to depart.

22.5kPa: Both Masters to decide if it would be safer to disconnect.

When at anchor or adrift, the Masters can also decide to direct both vessels in a position and a direction relative of the prevailing wind, where it would be safe to perform a controlled venting.

With these guidelines, the transfer operation will take place with sufficient margins against the cargo containment system pressure safety system settings.

Boil-off Management Equipment Failure

As described above, the saturated vapour pressure on the STBL should be such that even after the STS took place, the maximum saturated vapour pressure on the SS would remain well under safety valve settings.

If for any reason, the boil-off gas management equipment on either vessel should fail partially or completely, this should be reported as soon as possible to the other vessel. The cargo transfer should be discontinued if not already done by an emergency shutdown. If it is the boil-off gas management equipment of the STBL that failed, then the vapour manifolds should be shut to avoid increasing the pressure on the delivery vessel.

If the STBL has no longer control over its propulsion when adrift or at anchor, the SS should use the thrusters to direct both vessels in a direction where it would be safe for the STBL to perform a controlled venting operation.

If the boil-off management equipment on the SS should fail, the vapour manifold can be kept open so that the STBL can control the pressure on both vessels. If the boil-off management is insufficient to control this pressure, a controlled venting on the SS should be prepared.

If the SS can no longer use its engine, the other vessel should use her engines to direct both vessels in a direction where it would be safe to vent, and manifolds should be shut to avoid reaching high cargo tank pressures on either ship. The vessels should unmoor if the STBL is unable to keep both vessels in a favourable direction.

Maximum Levels

Since swell and weather might cause sloshing, care has to be taken not to activate the VHL alarm (98.5%) on the SS which causes the closing of the liquid filling valve or the EHL alarm (99%) which will cause an ESD. A topping-off rate of 500 - 800m³/h agreed at the pre-transfer meeting with a sufficient stagger between the levels of the tank shall suffice to counter the problem. The SS shall duly advise the STBL to reduce rates for topping-off.

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6.4 HOSE DISCONNECTION

Hose Draining and Purging

Once the transfer operation is completed, the manifold ESD valves on the SS are to be kept open initially. On the STBL, the manual double-shut valves will be closed whilst the ESD valves shall remain open, allowing the draining of manifold vertical riser. All hoses shall be drained from STBL towards the SS.

The proper draining shall be done by repeatedly pressurising the hoses using nitrogen till about 4-5 bar and flushing the LNG by opening the ESD bypass valve on the SS until the hoses are completely liquid free. Simultaneously, sea water spray on the hose's bight shall take place for de-icing and speeding up the liquid freeing. Both vessels have to carry out this operation together.

Hose Disconnecting

The ESD cable and the break away or trigger wire will be disconnected prior to disconnection of the cargo hoses to avoid entanglement. After disconnecting, each hose flange is to be blanked prior transportation. Manifold bypass valves on the SS to be cracked open to release any build-up of pressure in the cargo liquid hoses.

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7.0 Unmooring

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7.0 UNMOORING

Unmooring Procedures

The departure of the two vessels should undergo a strict regime of checks and preparation, similar to those taken before coming alongside. This operation remains a critical step, as both vessels are in close proximity of each other and as the displacement of each vessel has changed, this will affect the handling of each vessel directly.

Unmooring While One Vessel is at Anchor

Where STS transfer operations have taken place at anchor, under normal conditions of current and weather, unmooring may be carried out with the constant heading vessel remaining at anchor.

However, if the judgment of the person in overall advisory control, weather and current conditions so require, the constant heading vessel should weigh her anchor and unmooring should be carried out while making way.

Unmooring After Under Way Transfer

Where STS transfer operations have taken place while under way, it is normal to bring the combined two-vessel system head to the wind before unmooring, unless local conditions dictate otherwise.

Unmooring Checks

Prior to unmooring, it is necessary to complete the following checks to ensure that the vessels are fully prepared to depart without delaying the actual critical manoeuvre of both the vessels separating.

Side-to-Side Operations

Sufficient crew should be allocated to unmooring stations and consideration should be given to the following points:

- Side of cargo transfer cleared of obstructions including cranes.
- Method of disengagement and of letting-go mooring lines agreed.
- Fenders, including their towing and securing lines, checked to be in good order.
- Winches and windlass ready for immediate use.
- Rope messengers and rope stoppers ready at all mooring stations.
- Fire axes or other suitable cutting equipment available at each mooring station.
- Communications confirmed between vessels.
- Communications established with mooring gangs.
- Mooring gangs instructed to let-go mooring lines only when directed.
- Vessel traffic in the vicinity checked.

Unmooring Operation

Unmooring can be achieved by singling up fore and aft, then letting-go of the remaining forward mooring and allowing the bow to swing away from the constant heading vessel to a suitable angle, at which time the remaining stern mooring line is let-go and the manoeuvring vessel moves clear.

After disengaging, neither vessel should attempt to steam ahead or fall astern of the other, until both vessels are well separated.

It should be noted that local conditions or vessel configurations may cause difficulties in separating the two vessels and other alternative plans should be considered.

Because of the unpredictability of, and difficulty in assessing some conditions, it is recommended that unmooring at anchor be carried out only by persons with considerable experience in STS operations.

Unmooring Using Quick-Release Apparatus or Toggle Pins

Special care should be taken to letting-go the last lines, which should be released with great care. This operation should be planned in advance, be undertaken by an experience crew and requires good communication and supervision.

Different methodologies can be applied by STS operators and vessel's crew to carry out this task safely and effectively. One such method involves the use of quick-release hooks secured around the mooring bitt, or a toggle pin that is used in conjunction with a messenger to take the load of the mooring line while it is removed from the mooring bitt.

8.0 **Transfer Equipment**

8.1 **Cargo Transfer Equipment**

8.2 **Mooring Equipment**

8.3 **Personnel Transfers**

PRIME PURITY - STS OPERATING MANUAL

8.0 TRANSFER EQUIPMENT

Fenders

The provision of STS fendering equipment is currently detailed in the industry guideline, namely the ICS/OCIMF Ship to Ship Transfer Guide (Petroleum, Chemicals and Liquefied Gas).

Primary and secondary fenders should be of the pneumatic type, manufactured, tested and maintained in accordance with ISO Standard or other relevant ISO Standard. The requirements for fender numbers and sizes are detailed in tabular format in the OCIMF/ICS guide, which contains a strong recommendation that individual manufacturers or STS service providers are consulted to ascertain the required size and number of fenders for a particular operation.

As part of the vessel compatibility study, particular attention should be made to the parallel mid body length of both vessels. As the parallel mid body of the larger LNG vessels tends to be much shorter than that of a conventional oil tanker of similar length overall, the correct placement of the fender string is critical to ensure adequate hull protection for both vessels. This may result in the need for the fender connection wires to be made to specific lengths for the operation.

Fenders may be placed on either vessel based upon review of the available mooring points and fairleads to accommodate the fender towing wires. A suitable number of bitts and fairleads should remain available for the moorings when the vessels are alongside.

The fender towing wires (one forward and one aft) may be secured to a winch-mounted wire, if available, or made fast to a mooring bitt. If the fender towing wire is shackled onto a winch-mounted mooring wire, the winch brake should be correctly applied so as to avoid rendering, resulting in the fender string moving out of position.

Before mooring operations commence, proper securing of the fixed (hanging) fenders must be checked in advance. Rigging of fenders is a very dangerous operation and a risk assessment has to be done before operations commence. Every crew member involved in rigging must be instructed with a clear explanation of his personal and all team duties and the responsible officer must ensure full understanding of mentioned operation. Crew must be instructed about safety precautions as well as stressing extreme precautions during operation. The correct personnel protective equipment must be worn.

While rigging the aft fenders, the main engine may be be stopped down to avoid catching any floating ropes in the propeller. The officer of the watch (OOW) must be fully aware about principals of the main engine propulsion and trained and able to react accordingly in case of necessity.

Rigging must always be supervised by a responsible officer who must conduct it as per plan and maintain safe conditions at all times. Operations must be stopped immediately if in any doubt about the safety of the crew and should not be resumed until safe conditions are restored.

Maintenance and strict checking of mooring arrangements and fenders accessories should be performed on a weekly basis. Any shackles, wires and ropes with signs of substantial wear and tear must be replaced in advance. Vessel speed with rigged fenders should be at a minimum.

Primary Fenders

Primary fenders should be positioned with one at each end of the parallel mid body, with the remaining fenders placed forward and aft of the manifold. Fenders should be kept clear of the manifold area in case of a LNG release and so as not to pose a potential hazard to the cargo transfer equipment. Bitts recessed in the shell plating should not be used for securing fenders due to access and safe working load restrictions, and fenders should be kept clear of any recessed bitts and hull pilot access, if fitted, to prevent entanglement.

Secondary Fenders

Secondary fenders should be placed in order to protect bow and stern plating from inadvertent contact during mooring or unmooring. These should be identified as part of the ship compatibility review process and may be relatively high up on the vessel due to the flare of the bow and stern. Due care and attention should be made with regard to the location of suitable securing points as the preferred positioning of the secondary fenders may not be adjacent to a bollard or fairlead. Portable chocks may be supplied by the STS service provider for positioning the secondary fenders. In general, the SS is the smaller vessel and has the means to correctly hang the baby fenders.

The POAC should confirm the height of secondary fenders before the operation commences. During the course of the operation and changes in freeboard, the height of the secondary fenders should be adjusted as required. Also, frequent checks of the fender towing wires should be undertaken and chocks to be kept well greased. There may be occasions when the length of the fender towing wires may have to be adjusted to account for changes in freeboard of the vessel.

PRIME PURITY - STS OPERATING MANUAL

8.1 CARGO TRANSFER EQUIPMENT

Introduction

Before STS operations commence, the selection of equipment to be employed needs to be considered carefully. All elements of the cargo transfer system, including the loadings imposed on manifold working platforms, flanges, cargo transfer lines and their support arrangements should be fully evaluated, certified and shown to be fit for purpose in this application. Equipment includes the following:

- Connection pieces and reducers.
- Cargo transfer hoses.
- Hose saddles.
- Insulation flanges.
- Emergency release coupling.
- Cargo transfer hard arms.

Connection Pieces and Reducers

The vessel is designed with four liquid LNG connections and one vapour connection. Connection pieces can be attached to the liquid lines to act as reducers if smaller hoses are being used for the transfer. ‘Y’ connection pieces can also be connected to the liquid lines, allowing two hoses to be used for one line.

Hose Saddles

Hoses normally cross over the vessels side rails via saddles which are designed to support the hoses during transfer operations.

Cargo Transfer Hoses

Transfer hoses should be of sufficient length to avoid over-stressing and chafing throughout the cargo transfer. To establish the correct length of hose, changes to the relative freeboard and vessel’s movements must be taken into account, as well as the minimum allowable bend radius of the hose, the horizontal distance between vessels, any differences in manifold alignment, any limitations imposed by the vessels equipment and any other special characteristics of the two vessels involved.

The hoses used for the STS transfer should be specially designed and constructed for the product being handled and the purpose for which they are being used. Hoses used should comply with EN1765 (or latest equivalent) with regard to specification for the assemblies and with BS1435 (or latest equivalent) and OCIMF guidelines with regard to their handling, inspection and testing.

Hoses should bear the following durable indelible markings:

- The manufacturer’s name or trademark.
- Identification of the standard specification for manufacture.
- Factory test pressure (equal to rated working pressure, maximum working pressure, maximum allowable working pressure).
- Month and year of manufacture and manufacturer’s serial number.
- Indication that the hose is electrically continuous or electrically discontinuous, semi-continuous or anti-static.
- The type of service for which it is intended, eg, LNG.

Test data with respect to each hose should be available and should be sighted prior to the hose being used for transfer.

Hoses should be withdrawn from service and retired against defined criteria which may include the following:

- The presence of defects detected during visual inspections. Defects prompting retirement could include irregularities in the outside diameter, such as kinking, damaged or exposed reinforcement or permanent deformation of the casing and damage, slippage or misalignment of end fittings.
- After a defined period in service, established in consultation with the manufacturer.
- When the temporary elongation of the hose, measured during routine pressure tests, exceeds maximum allowable values.

A visual inspection of each of the hose assemblies should be carried out before they are connected to the manifolds to determine that they are free of damage.

If damage to a hose or flange is present, the hose should be withdrawn from use for further inspection, repair or retirement.

The maximum permissible flow velocity through a hose is limited by the construction of the hose. The hose manufacturer’s recommendations and certification should give details.

STS transfer operations require hose connections to be well made. Flanges or, if used, quick-release couplings, should be in good condition and properly secured to ensure leak-tight connections.

In order to eliminate the potential for incendive arcing between the two vessels when presenting the hose string for connection:

1. A single insulating flange should be fitted within each hose string, or at each vessel’s manifold.
2. One length of of electrically discontinuous hose should be fitted in each hose string.

To prevent localised accumulation of electrostatic charge in a hose, the string should be electrically continuous on both vessels on either point of the insulation. This means, that when an insulating flange is used at the manifold, it is only used on one vessel.

Insulation Flanges for Transfer Hoses

Electrical arcs may occur when connecting and disconnecting the cargo hoses. The cargo hose provides an electrical path between the two vessels.

Electrical current will flow through this path due to the differences in potential between the two vessels.

As well as both vessels switching off there cathodic protection system prior to arriving alongside each other, electrical discontinuity in the cargo hose can be introduced by means of an insulating flange fitted between the cargo hose flange and manifold flange. This is effective in eliminating the current and any sparking.

Care should be taken to apply correct force with the torque wrench during the connecting procedure in order not to break the PTFE washers.

Emergency Release Coupling

An ERC (Emergency-Release Coupling) can be connected to each hose on the vessels manifold. Excessive movement at the manifold will activate the ESD device which is a simple wire attachment that will operate the hydraulic power release, when the wire reaches the set physical limit.

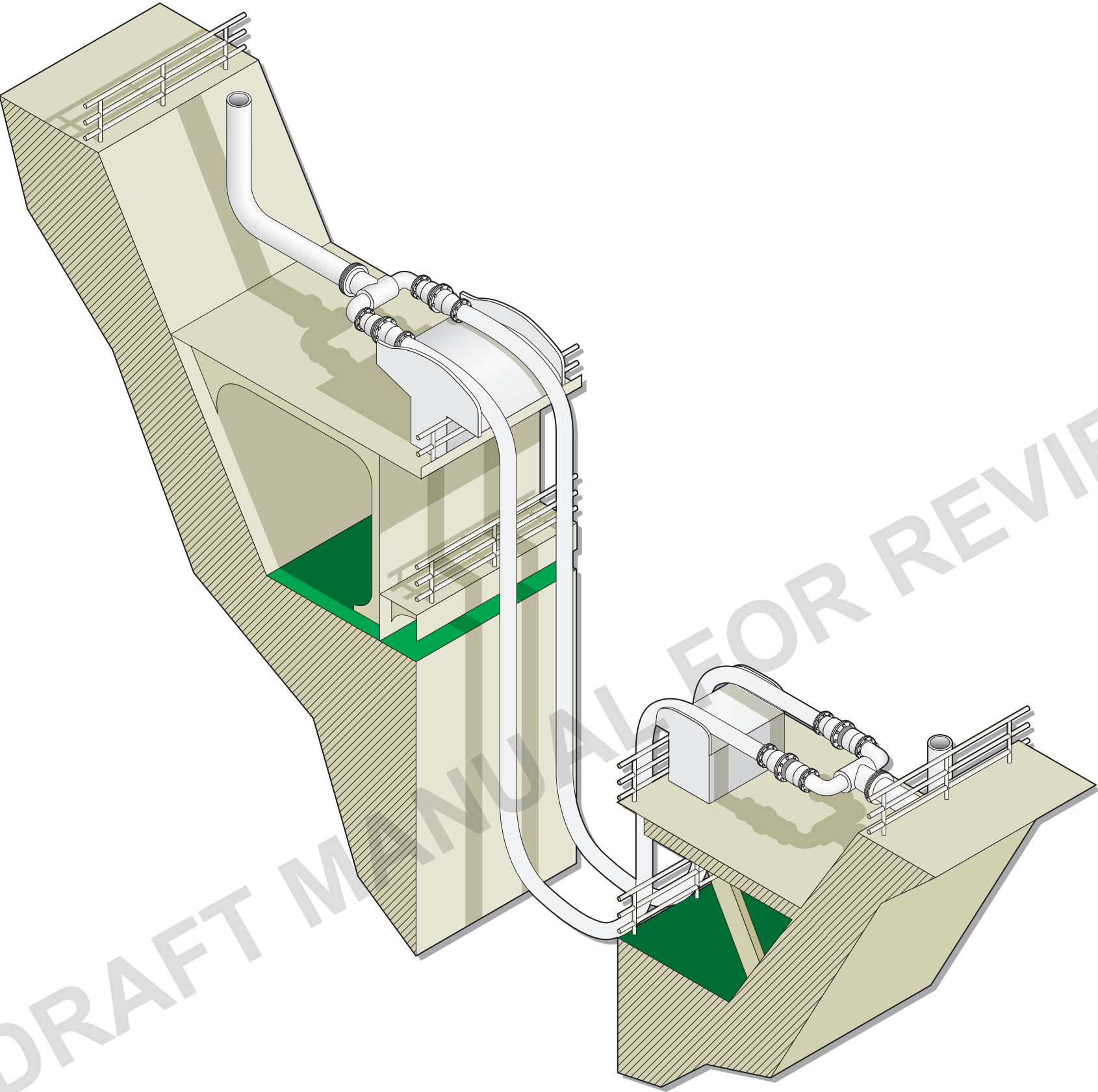
Cargo Transfer Hard Arms

Where hard arms are considered for use in LNG STS transfer operations, the following factors should be taken into consideration:

- Accelerations.
- Permissible manifold loadings.

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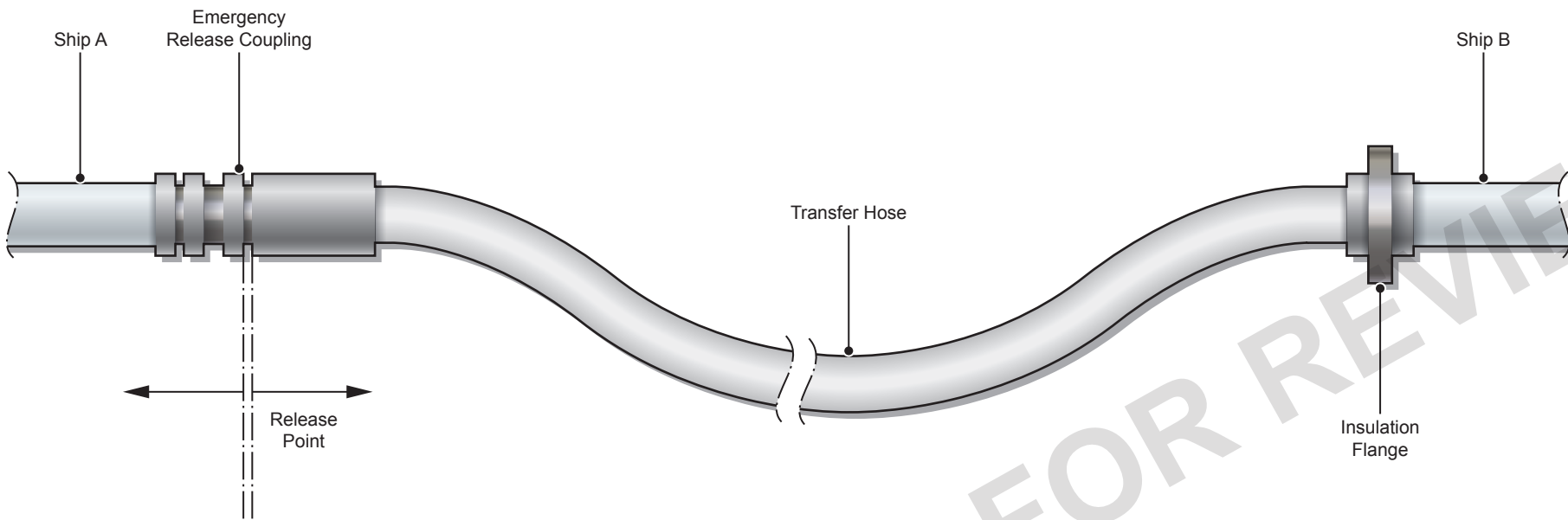
Illustration 8.1.1a Typical Transfer Hose and Y Piece Arrangement



PRIME PURITY - STS OPERATING MANUAL

- Arm support arrangements.
- The effect of vibration on the arm.
- Size of the arm.
- Connectability.
- Vertical and horizontal vessel movement.
- Allowable flow velocity.

Illustration 8.1.1b Insulation Flange Arrangement



PRIME PURITY - STS OPERATING MANUAL

8.2 MOORING EQUIPMENT

Moorings

A detailed mooring compatibility study shall be carried out in the preparation phase prior to the STS operation. The mooring studies should also be validated by means of appropriate mooring software. Due regard should be paid to the guidance in the following publications:

- OCIMF Mooring Equipment Guidelines.
- International Safety Guide for Oil Tankers and Terminals.
- In the ICS/OCIMF Ship to Ship Transfer Guide (Petroleum, Chemicals and Liquefied Gas).
- General comment, consider inserting publishers of above documents.

Ancillary Equipment

All ancillary equipment, such as wires, messengers, stoppers, strops and shackles should be certified and inspected for condition prior to being used during the STS operation.

8.3 PERSONNEL TRANSFERS

General

In general, it is recommended that the transfer of personnel between ships is kept to an absolute minimum. As a rule, only crew from the SS should be transferred between the two vessels.

Personnel may be transferred utilising a Billy Pugh personnel transfer basket between the vessels. The crane utilised to lift this basket must be suitable for lifting personnel. Some vessels have their cranes certified by class for lifting personnel (basis off-shore rules) even though no specific rules apply to this in shipping. In case no certification has been obtained, a separate evaluation in the form of a formal risk assessment may find the cranes nonetheless suitable for transfer of personnel if, and only if, no unacceptable safety issues have been identified. The criteria when using cranes for personnel transfer depend on suitability of cranes, acceptance of vessel operator, permission by local authorities.

In case cranes are found not suitable, or if otherwise requested or preferred by local administration, personnel may be transferred between vessels by means of a boat using a suitable pilot ladder or combination of pilot ladder and accommodation ladder. It is anticipated that the environmental conditions determining the STS operability window will usually also allow for personnel transfer by boat. For further information on personnel transfer procedures equipment, reference should be to the ICS/OCIMF Ship to Ship Transfer Guide (Petroleum, Chemicals and Liquefied Gas).

9.0 Emergency Planning

9.1 Emergency Disconnection

DRAFT MANUAL FOR REVIEW & COMMENT

PRIME PURITY - STS OPERATING MANUAL

9.0 EMERGENCY PLANNING

General

The LNG STS transfer emergency procedures are drawn up as a supplement to the operator’s existing Emergency and Contingency Manual.

In the event of an emergency, the very first action shall be to stop the cargo transfer, ie, stopping the main cargo pumps in order to stop the flow of cargo before actuation of the ERC. In the case of vessel’s breakaway, the break away or trigger wire will initiate ESD1. Once cargo pumps have been stopped and the manifolds closed, ESD2 will be activated which releases the ERC.

At all times, a seaman is on standby at the manifold and a qualified officer is on duty. The qualified engineer or officer will be fully conversant with the operation of the ERC system (the operation of the ERC takes place from the service ship). Regular checks of the HPU unit are done and recorded by the cargo engineer, to assure its proper functioning in case of need. In case of a significant problem with the ERC or HPU, STS transfer will be stopped. The operation shall resume only when the complete release system is restored to proper operation. The ERC system, by design, is powered by a fail-safe hydraulic accumulator system that allows actuation of the ERC in the event of a power failure.

In addition to the permanent presence of personnel near the manifold, remote observation of the cargo manifold area and ERC system may be available from the cargo control room and bridge by means of the CCTV cameras on board the SS.

Mooring Failure

In the event of a mooring line failure, Masters shall evaluate the situation and the cargo transfer operation may be suspended until a safe mooring condition is verified. The suitability of the mooring system, including redundancy of lines shall be confirmed during the STS ship compatibility and mooring study.

Adverse Weather

In case of deterioration of the weather conditions, the STS transfer operation shall be suspended in due time so that preparations can be made to disconnect cargo hoses, separate the vessels and allow both vessels adequate time to resolve cargo tank partial filling conditions in a controlled fashion. It is incumbent upon the Master of each vessel to decide when to suspend the STS operation on the basis of his vessel’s ability to manage partial filling conditions and to prevent damage to the cargo containment system due to sloshing liquid impacts.

Prior to conducting the STS operation, the cargo transfer shall be planned to prevent exposure of both vessels to sloshing damages. The actual cargo transfer sequence shall be part of the site specific STS risk assessment process prior to the STS operation. If necessary, both ships shall transfer cargo internally in accordance with predefined schedules and in accordance with GTT guidelines.

9.1 EMERGENCY DISCONNECTION

General

In situations where it is deemed necessary for both vessels to depart from each other instantly or where the vessels are experiencing an unexpected separation, the cargo transfer hoses will be disconnected by means of the ERC system.

After an emergency shutdown (ESD) is actuated, the ERC system on board the SS activates and releases the STS hoses. The hose brake system in the saddles will ensure the hoses are lowered in a controlled manner. When all hoses are lowered, the air supply has to be opened to the declutching system. This action will release the rope that controls the lowering and both vessels will be ready to depart.

The hoses remain thus on the delivery vessel, and any possibility of liquid block in the hoses shall be managed by opening the manifold cooldown/bypass valves on the STBL if available. The main cargo transfer manifold (ESD valves) valves shall be kept shut. This will allow any vapour generated inside the hoses to be safely evacuated towards the cargo tanks. The hoses will still have to be liquid freed, de-iced and purged before it is deemed safe to recover and eventually reconnect them.

In the case of an emergency where a breakaway is the preferred response, the estimated time required to conduct the operation is as follows:

ESD activation/securing cargo transfer:	60 seconds
Hose disconnection:	60 seconds
Emergency unmooring:	13 minutes
Total estimated time for vessel breakaway:	15 minutes

Emergency Escape

The primary means of escape from one or both vessels involved in the STS operation in case of an emergency is each vessel’s outboard lifeboat or shore gangway. As a secondary means of escape, the ship’s gangway and pilot ladder can be used to disembark to the lightering support vessel.

Local escape and evacuation from the STBL cargo manifold area is provided with a temporary ladder(s) lashed in place in between cargo manifolds which allows cargo watch personnel to escape from the manifold area without having to climb under or over cargo transfer hoses and equipment. The SS may be fitted with permanent additional manifold escape routes which precludes the use of temporary ladders.

Ice Accumulation

As part of industry standards and cryogenic equipment type approval, the equipment has been designed for up to 1”/25mm of solid ice accumulation (not snow or frost).

The cargo transfer hoses and ERCs have been type-approved for use in LNG STS operations with this standard in mind. Current experience proves the transfer hoses and ERCs are not affected by this amount of solid ice built up in a variety of climatic conditions. However, during the STS transfer operation, the responsible officer shall closely follow the ice formation on the STS transfer hoses. In case of ice formation approaching the above limits, ice removal methods commonly used in land-based terminals, such as water spray, shall be used to limit or remove the ice. In cases where ice build-up is excessive, the cargo transfer shall be reduced or stopped and the transfer hoses and/or ERC will be de-iced prior to resuming the operation.

Dead-Ship Emergency Release System Activation

To ensure that the mechanical integrity of the transfer system is not compromised during emergency scenarios, the release system is able to function under a dead-ship condition (ie, no motive power is available from the ship systems). In addition, the ERS is capable of releasing the transfer lines automatically before their safe working envelope is exceeded.